

Stretch Your Skill: Co-Embodiment Training on Motif-Matched Beat Saber Replays

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Abstract

We present a data-driven system for personalized motor skill training in virtual reality (VR). The system implements co-embodiment training in the popular VR rhythm game “Beat Saber,” where learners are paired with virtual teachers selected from real player recordings based on “motion motifs,” recurring patterns in a player’s movements. We analyzed hundreds of game replays and found that as players improve, their motion motifs gradually diverge, suggesting that exposure to unfamiliar yet attainable movement styles support skill acquisition. We hypothesize that the ideal teacher matches the learner’s skill level but employs different motion motifs. We conducted a user study (N = 15) to evaluate this strategy against baseline conditions. Results show that while similarly skilled teachers improve learner’s sense of agency, our proposed method significantly reduces physical learner-teacher distance, indicating more effective and synchronized learning. This work demonstrates an innovative VR system that leverages community data for personalized and practical motor skill training.

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1 Introduction

Virtual Reality offers powerful new paradigms for motor skill learning (MSL). One promising method is co-embodiment, where a learner and teacher share control of a virtual body to transfer complex skills [Fribourg et al. 2020; Kodama et al. 2023]. However, the critical “teacher” role requires an experienced human instructor collaborating in real-time, or, more recently, an AI trained on such instruction data [Takita et al. 2024].

This study introduces a novel learning experience that leverages the vast repository of existing player recordings from the popular VR game **Beat Saber**—a domain that involves complex eye-hand coordination and upper-body motor skills—to create personalized co-embodiment training. We treat the millions of available game replays [Nair et al. 2023] as a library of digital teachers, each with a unique movement profile. Our system enables learners to match

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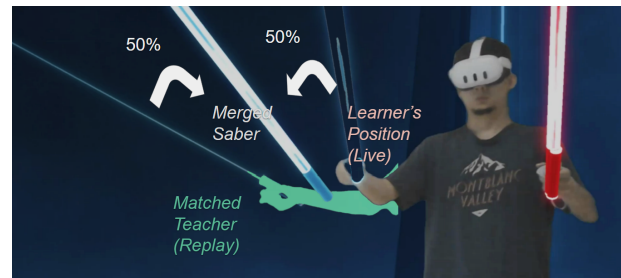


Figure 1: The co-embodiment experience in Beat Saber. The virtual sabers’ motion is a blend of the user’s controllers and the selected teacher recording. In-game visuals (c) Beat Games

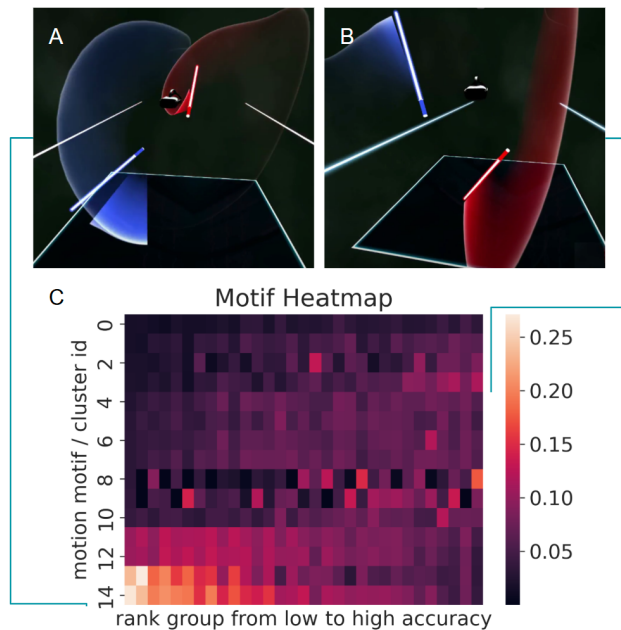


Figure 2: A: Example motif in novice group (e.g. unnecessary extra swings near the end of one cut); B: Example motif in expert group (e.g. clean and decisive cuts); C: The usage of different motion motifs as we move from novice groups (left) to expert groups (right, each vertical stripe one player group and brighter color means more usage of that motif)

and merge with the most suitable teacher for their own profile (Fig.1). The core of our work is a data-driven method for selecting the right teacher to help a player improve.

2 System Design

Our system is founded on a key insight from analyzing real player data, which informs the design of our co-embodiment experience.

To determine what makes a good teacher, we first analyzed how players' movements change as they improve. In the game of Beat Saber, players swing two light sabers to cut incoming blocks at the proper timings, in sync with a song. Beat Leader, a community leaderboard, hosts millions of such head and hand movements for competition and rankings. We processed 652 replays from one moderately challenging song¹ using a variational autoencoder [Luxem et al. 2022] to identify recurring movement patterns. We grouped the encoded vectors of short motion segments into clusters, whose distribution reveals a player's "motion motif," or recurring movement blocks [Aristidou et al. 2018]. The analysis revealed a clear trend: as players get better, their repertoire of motifs becomes more diverse and unique (Fig. 2C shows the usage ratio for each motif across skill groups). While novices share a common, limited set of moves, experts diverge, each developing their own signature style. This suggests that improvement comes not from mimicking one perfect form, but from exploring and integrating movements new to the learner.

3 User Study

Based on this insight, we built a training system as a mod for Beat Saber where the user's sabers are blended 50/50 with a teacher's replay (Fig. 1). This creates the experience of sharing 50% control of one's sabers with a ghost teacher, subtly being guided by the teacher's movements. We designed a user study (N=15; 10 male; 9 new to Beat Saber, 4 new to VR) to compare the co-embodiment experiences from three teachers:

- **C1 (The Pro):** Merging with the #1 ranked player on the given song. A common but often overwhelming baseline.
- **C2 (The Twin):** Merging with a player of similar skill level (from a pool of five players scoring just above the learner) and the most similar movement style (smallest distance between average motion vectors from autoencoder). This partner feels familiar but may offer little new information.
- **C3 (The Stretch):** Proposed method. Merging with a player of similar skill level (from the same pool as C2) but the most *dissimilar* movement style (i.e., the largest distance). This partner offers unfamiliar movements at a comparable skill level, "stretching" the user's existing movement patterns.

The order of conditions was counterbalanced using a Latin square design. For each condition, participants played a medium-level 1 min 45 s song² once without a teacher (pre-test), then once with the co-embodiment teacher for training, and finally once more without the teacher (post-test). We measured the change in score between the post-test and pre-test. We also evaluated kinematic data (reduction in learner-teacher hand distance, or L-T distance), and subjective questionnaires [Roth and Latoschik 2020].

While we found no significant difference in score improvement in the short session, the kinematic data and subjective ratings reveal a clear advantage for our proposed method. Subjectively, we observe a significantly higher sense of agency for both the Twin (C2) and

the Stretch (C3) compared to the Pro (C1). Participants felt more in control and attributed the stretch to themselves more when the skill gap was manageable.

Objectively, the effect was captured by the L-T distance. In previous work, the hand distance between learner and teacher increases as the learner fights for control [Hagiwara et al. 2020; Kodama et al. 2023]. Here, only the Stretch (C3) caused a significant *decrease* in L-T distance, indicating better synchronization between learner and teacher.

Furthermore, anecdotal feedback indicated that participant learners could perceive distinct personas in the teachers: some "Pro" teachers felt too robotic for them, while our proposed ones felt softer and more human.

4 Demonstration

Our system transforms a popular VR rhythm game, Beat Saber, typically played solo, into a personalized duet by allowing learners to virtually merge with the embodied replay of another player's past recordings on the leaderboard.

At the demo booth, attendees will first play a short Beat Saber song, approximately 1 min long, to profile their motion motif usage. Based on that, our system will then select the best-matched virtual teacher for them to train with, as described in the system design.

They can also switch between the best-matched teacher and other teacher options in our co-embodiment system, thus directly feel the difference in guidance, synchrony, and their sense of control. This allows them to experience firsthand how a well-matched teacher, selected by analyzing community data, makes learning a complex skill feel more intuitive and natural.

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¹<https://beatleader.com/leaderboard/global/4a2a91/>

²<https://beatleader.com/leaderboard/global/4adb51/>